ROMERO ALEJANDRO

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I design resilient systems and engineer them from the metal up.

From low-overhead Rust LLM streaming runtimes (70B models at 1.1GB RAM) and production healthcare backends (11,133 records migrated) to autonomous robotics and synaptic graph memory. 10+ projects engineered. Based in Japan.

11,133	1.1 GB	301	Lv. 63
RECORDS MIGRATED	RAM FOR 70B MODEL	RUST TESTS PASSED	SIGNATE CLOUD (879+ EX)

CORE FOCUS & ENGINEERING PILLARS

PILLAR 01 — Arch Linux • Rust Multi-Crate

Production Systems & Arch Linux Infrastructure

Operating on Arch Linux as daily driver. Engineered multi-crate pure Rust workspaces, high-concurrency Axum/Tokio web backends, and low-latency background daemons. Demonstrated zero-loss data migration of 11,133 patient records from legacy Access to modern SQLite, containerized with Docker/Podman.

Arch Linux | Rust Multi-Crate | Docker/Podman | Systemd Daemons | Zero-Loss Migration

PILLAR 02 — Robotics Teaching • Multicultural Education

Robotics Education & Global Event Leadership

Experienced educator and community leader. Led and taught hands-on robotics, programming, and maker workshops to children and adults of diverse multicultural backgrounds across multiple events. Directed and coordinated week-long technical events and humanitarian initiatives across 3 continents.

Robotics Teaching | Multicultural Education | Week-Long Event Management | 3 Continents | Team Leadership

PILLAR 03 — Raspberry Pi 5 • Raspberry Pi Zero 2W

Single-Board Computers & Microcontrollers

Extensive hands-on engineering across single-board computers and microchips: from Raspberry Pi 5 and Pi Zero 2W to Orange Pi 6 Plus and ESP32 chips (ESP32-S3 Sense, ESP32-C3). Proficient in hardware interfacing, low-latency firmware, I2C/SPI protocols, and edge Linux optimization.

Raspberry Pi 5 | Raspberry Pi Zero 2W | Orange Pi 6 Plus | ESP32-S3 Sense | ESP32-C3 | Embedded Hardware

PILLAR 04 — 3D Printing • CAD Design

Self-Taught 3D Printing & Rapid Prototyping

Self-taught in 3D CAD modeling, slicing, and additive manufacturing. Capable of designing custom robotic chassis, sensor mounting brackets, and custom device enclosures, iteratively testing and refining physical prototypes from raw ideas into functional hardware.

3D Printing | CAD Design | Iterative Prototyping | Hardware Fabrication | Enclosure Design

PILLAR 05 — 3D Galaxy Animation • Game Development • UX/UI Architecture

Interactive 3D Systems, Animation & Game Design

Merging visual aesthetics with low-level computation. Engineered interactive 3D visualizers like the Dendrite 3D Orbital Galaxy Memory Atlas in Cynapse, crafted unique modern user interfaces, and combined design architecture with programming logic to develop complete games.

3D Galaxy Animation | Game Development | UX/UI Architecture | Figma | Interactive Prototyping

FLAGSHIP SYSTEMS 01 — 03

Low-overhead Rust inference runtimes, graph memory, and zero-loss production database modernizations.

PROJECT 01 https://github.com/Alartist40/cynapse.git

Cynapse

High-performance, offline-first AI agent platform engineered in 100% pure Rust. Features the 4-tier Dendrite SQLite FTS5 graph memory, Leafcutter tensor execution core, 3D orbital galaxy visualization, and an interactive retro TUI harness. 301 verified tests passing.

KEY ARCHITECTURAL HIGHLIGHTS

- 100% Pure Rust Multi-Crate Architecture: Decoupled modular workspace comprising cynapse-harness, cynapse-tui, cynapse-core, cynapse-memory, cynapse-engine, and leafcutter_core for blazing speed and zero memory leaks
- Dendrite 4-Tier Synaptic Graph Memory: SQLite FTS5 full-text indexing + BM25 relevance ranking that organically forms neural connections between concepts, sessions, and facts
- Unified Hybrid Multi-Engine: Native Leafcutter Rust tensor core for raw GGUF & Safetensors execution combined with Tier-1 llama.cpp / Ollama high-speed streaming

Rust | AI Agent | Graph Memory | TUI | Offline-First

PROJECT 02 https://github.com/Alartist40/LeafcutterLLM.git

LeafcutterLLM

Memory-first adaptive LLM runtime written entirely in 100% Rust. Runs 1.5B to 70B parameter models on commodity CPUs without discrete GPUs via 3-tier streaming execution, achieving 1.1GB peak RAM on Llama-3.1-70B and 21.6x speedup over AirLLM.

KEY ARCHITECTURAL HIGHLIGHTS

- Pure Rust Native Engine: High-efficiency runtime written from scratch in 100% Rust — zero Python, zero CUDA, zero external runtimes
- 3-Tier Adaptive Memory System: Tier 1 (Full RAM/VRAM cache), Tier 2 (Cached streaming with prefetch), and Tier 3 (Low-RAM streaming using madvise(DONTNEED) layer eviction for tight memory limits)
- Validated 1.5B to 70B Model Roster: Verified coherent reasoning on Ornith 1.0 9B/35B MoE, Qwen3.8-27B, Meta-Llama-3.1-70B (40.3GB file executed at 1.1GB peak RAM), Ministral-3B, and Qwen2.5-1.5B

Rust | LLM Engine | SIMD | Open Source | Sub-1GB RAM

PROJECT 03 https://github.com/Alartist40

E&P Medical Management System

Production healthcare practice management platform deployed at E&P Medical Centre in Botswana. Fully replaced a legacy 96MB Microsoft Access database, flawlessly migrating 89 tables and 11,133 patient records to a zero-dependency Rust and SQLite single binary.

KEY ARCHITECTURAL HIGHLIGHTS

- Production Pure Rust 1.97+ Web Backend: Built with Axum, Tokio async multi-threaded runtime, and Serde, completely eliminating Python runtime overhead and external daemon dependencies
- Flawless Zero-Loss Data Migration Pipeline: Extracted 89 tables and 11,133 relational patient records from legacy 96MB Access database (SURGIC.mdb) into structured, portable JSON datasets and verified relational SQLite tables
- Static Rusqlite Storage Core: Inlined zero-dependency SQLite engine (epmed.db) with WAL journal mode, providing instant multi-token patient search with zero latency

Rust | SQLite | Healthcare Production | Data Migration | Axum

FLAGSHIP SYSTEMS 04 — 07

Autonomous robotics, LoRa off-grid mesh, SBC full-text search, and local-first spaced repetition.

PROJECT 04 Alartist40/the-pathfinder-eye

The Pathfinder Eye

Autonomous robotics platform with native Go/Rust brain, omnidirectional mecanum locomotion, face recognition security authority, and real-time YOLO vision at 30fps with multimodal AI reasoning on Raspberry Pi 5.

- Zero-Python Architecture: Fully native Go/Rust stack eliminates Python overhead, reclaiming 400MB RAM and 1.2GB storage
- Cynapse Neural-Link Memory: Graph-based knowledge engine connecting [[People]], [[Roles]], and [[Events]] with bi-directional backlinks

Rust | Go | Robotics | Computer Vision | Raspberry Pi 5

PROJECT 05 Alartist40/paraclea.git

Paraclea

Pure Rust offline AI companion and intelligent librarian engine. Features personalized Dendrite v2 knowledge graph memory, 219 Bible translations across 30 languages, 211-chapter spiritual/survival/medical library, and Reticulum LoRa off-grid mesh communications.

- Pure Rust AI Companion & Librarian Architecture: Built entirely in pure Rust for blazing fast offline text search, multi-translation verse comparison, and intelligent recall
- Personalized Librarian AI & SOUL System: Managed by a tiny, adaptive local AI acting as a dedicated librarian with customizable personality profiles (SOUL.md, IDENTITY, TOOLS) that tune to individual communication styles

Rust | AI Librarian | Knowledge Engine | Reticulum Mesh

PROJECT 06 Alartist40/mazzaroth

Mazzaroth

Pure Rust offline digital knowledge vault and source-verified research engine for single-board computers. Delivers sub-millisecond BM25 full-text indexing, offline vector search, 62 countries of geopolitical maps, and an interactive 1,105-node 3D knowledge graph.

- 100% Pure Rust Multi-Engine Vault: Single static binary engineered for ultra-low resource single-board computers (Raspberry Pi, Orange Pi) with zero external runtime dependencies
- Sub-Millisecond BM25 Retrieval Engine: Inverted index and lexical scoring running over thousands of historical, medical, and survival source documents without network access

Rust | Offline Knowledge Vault | BM25 Search | 3D Graph | SBC

PROJECT 07 Alartist40/hikarigo

Hikarigo

Privacy-first, open-source language learning and cognitive study platform built in Go and clean HTML/CSS. Delivers zero-tracking SM-2 spaced repetition, instant vocabulary recall, and local-first data ownership with sub-10ms response times.

- Clean Go Micro-Service Architecture: High-concurrency backend written in Go, compiling to a tiny single executable with zero external server dependencies
- Privacy-First Local Storage Engine: Zero analytics, zero third-party telemetry, and local-first SQLite persistence storing spaced-repetition study history on the user's device

Go | Privacy-First | Educational Platform | Spaced Repetition | SQLite

TECHNICAL SKILLS MATRIX

Systems & Low-Level Architecture

Rust (primary language, 100% pure cores, multi-crate workspaces), Go, C/C++, SIMD (AVX2/NEON), arena memory allocation, zero-dependency single binaries, Arch Linux, Bash, Podman/Docker.

Edge AI & LLM Inference Runtimes

3-tier adaptive streaming runtime (70B LLM at 1.1GB RAM), GGUF/Safetensors execution, GBNF grammar sampling, SQLite FTS5 synaptic graph memory, BM25 text retrieval, Python (scikit-learn).

Hardware & Embedded Robotics

Raspberry Pi 5, Pi Zero 2W, Orange Pi 6 Plus, ESP32-S3/C3, I2C/SPI protocols, mecanum locomotion kinematics, real-time YOLO vision (30fps), 3D CAD modeling & rapid FDM additive prototyping.

Web Backends & Data Modernization

Axum, Tokio async runtime, Actix-web, embedded SQLite WAL mode, complete zero-loss legacy Access to SQLite migration (89 tables / 11,133 records), HTML5/CSS3, TypeScript/React, Git/CI/CD.

Language Proficiency (4 Languages)

Spanish (Native speaker), English (Business level / multinational education), Japanese (Conversational / technical coursework / JLPT N2 study), French (Basic study).

Certifications & Emergency Training

Generative AI Passport Certified (Oct 2025), Japanese Red Cross First Aid Certified (Tokyo Branch, Apr 2026), SIGNATE Cloud Level 63 (870+ exercises completed), Anthropic Claude 101 Certified.

ACADEMIC & PROFESSIONAL JOURNEY

2025 - 2027 — Chuo Computer & Communication College (中央情報専門学校) — Associate Degree / Senmonshi (専門士) in AI Systems

Expected graduation March 31, 2027. Specializing in computer science, low-level inference runtimes, edge robotics, and autonomous agent architectures. Full work visa sponsorship parity under Japanese Immigration Law.

2023 - 2025 — ISI Language School, Takadanobaba, Tokyo (ISIランゲージスクール 高田馬場校) — Japanese Language Immersion

Graduated March 2025. Intensive Japanese language training, intercultural communication, and professional conduct.

2020 - 2023 — Dalian Medical University (大連医科大学, China) — Clinical Medicine & Physiological Systems

Completed June 2023. Acquired deep clinical intuition in interconnected biological systems, pharmacology, and high-pressure emergency care.

2018 - 2019 — South Western University (Philippines) — Pre-Medical Curriculum

Undergraduate pre-medical coursework in Cebu before transferring to Dalian Medical University in 2020.

2014 - 2017 — Gaborone International School (Botswana) — Secondary Education Graduate

Multilingual international schooling foundation in Southern Africa.

2018 - 2023 — E&P Medical Centre (Gaborone, Botswana) — Medical Administration & Core IT Systems Modernizer

Handled clinical reception, triage intake, and spearheaded modernization of clinic infrastructure: successfully migrated 89 tables and 11,133 records from Access to an air-gapped Rust/SQLite single-binary system.

PERSONAL PURSUITS & SELF-DISCIPLINE

- **Sketching:** Self-taught since high school, continuously honing visual observation and analytical drawing.
- **Strength Training:** Daily home training regimens for physical stamina, focus, and self-discipline.
- **UI Design:** Rapid interactive prototyping, design systems, and vector logo design in Figma.

SPECIALIZED ACADEMIC SUBJECTS

- **AI Project Development:** Full-cycle implementation of edge LLM streaming runtimes and autonomous agents.
- **Computer Science:** Memory layout, arena allocation, SIMD parallelism, zero-cost abstractions.
- **Algorithms & Data Structures:** Graph traversal, SQLite FTS5 BM25 search, associative memory.

VOLUNTEER & HUMANITARIAN LEADERSHIP

Over a decade of hands-on youth technology education, international camp operations, and overseas medical triage and surgical support across 3 continents.

Community & Educational Leadership in Japan (Tokyo International Church & Beyond / 2025–2026)

- Planned and instructed classes in crafts, survival skills, filmmaking, programming, and robotics for children and adults
- Managed event planning and operations, youth camp coordination, and safety management
- Directed media and public relations: designed posters, video thumbnails, and website architecture
- Served as a bilingual English-Japanese interpreter supporting multi-generational and international community activities

Technical & Engineering Leadership (CCMC / 2025–2026)

- Led robotics architectural strategy and design decisions for the robotics development team at Chuo Computer & Communication College (CCMC)
- Supported and consulted on robotics manufacturing and coding projects across classes and departments
- Mentored classmates in software development workflows, tooling, and low-level engineering

Global Medical & Community Outreach (Botswana & Philippines / 2015, 2018–2019)

- Botswana (2015): Accompanied multidisciplinary medical team at age 15; conducted health checkups for 100+ community members including blood pressure, pulse, and ABO/Rh blood typing
- Leyte Island, Philippines (2018–2019): Trekked 6+ hours through mountainous terrain to deliver essential medicine to remote indigenous settlements; supported field consultations and care as part of medical team
- Mindanao, Philippines (2018–2019): Supported medical triage, vital sign monitoring, and diagnostic blood typing as part of mobile healthcare team
- Philippines (2018–2019): Clinical assistance in surgical suturing, wound dressing/gauze management, and circumcision surgical support (German and Philippine methods)
- Cebu, Philippines (2018–2019): Environmental cleanup and community beautification initiatives

HUMANITARIAN PHILOSOPHY & FIELD ADAPTABILITY

From clinical triage in Botswana at age 15 to 6-hour mountain treks delivering medicine to remote Philippine tribes, these experiences forged extreme resilience, emergency calm, and deep cross-cultural adaptability. Combined with bilingual interpreting and hands-on technology education across Japan, these humanitarian foundations drive my commitment to engineering systems that genuinely protect, serve, and empower communities worldwide.

Inquiries, Engineering Roles & Collaborations

Systems engineering, low-level Rust, LLM streaming engines, and robotics.

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